Lesson Notes

MODULE 8 | LESSON 3

REINFORCEMENT LEARNING FOR PORTFOLIO MANAGEMENT

Reading Time	30 minutes
Prior Knowledge	Gradient descent
Keywords	Agent, Actions, Environment, State, Observation, Reward, Policy, State-value function, Action-value function, Bellman equation, Policy gradient method

In the last two lessons, we focused on backtesting techniques to help us discover improved portfolio construction models. In this lesson, we continue our search for optimal portfolio construction with the help of reinforcement learning.

1. The Reinforcement Learning Model of the World

While it is assumed that you have already studied reinforcement learning, these lesson notes and the required readings for this and the next lesson do not necessarily require prior familiarity. The longest required reading, by Filos, explains the basics in enough, but not too much, detail. Indeed, the first three chapters (almost 40 pages) cover material that you have already seen: portfolio optimization, time-series analysis, performance and risk metrics (like the Sharpe ratio and Value at Risk), and general deep learning (like recurrent neural networks, or RNNs). That said, Filos' review is succinct and provides a mathematical formalism that will serve you well in the chapters that follow where reinforcement learning is introduced and explained.

On the surface at least, reinforcement learning (RL) is based on a very simple but very straightforward and realistic model. This model could be used very generally to describe or understand how individuals interact in their world. The model could also be used more specifically to show how portfolio management works. For example, let's say:

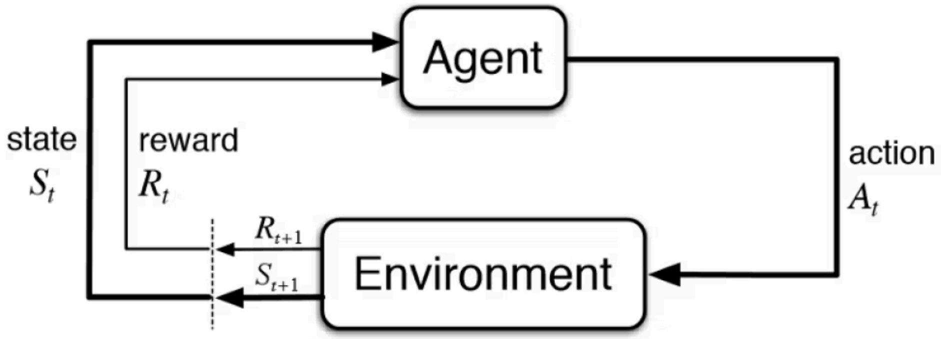
1. You are an equities portfolio manager.
2. This means you buy and sell stocks.
3. In this sense, you are a part of the market for stocks and a part of the financial system.
4. You decide which stocks to buy or sell by watching the market and/or the financial system.
5. When you do a good job of choosing which stocks to buy or sell, you get paid more.
6. You make these decisions based on your experience.

All of these matter-of-fact statements are easily mapped to the RL model of the world:

1. You are an **agent**, where "agent" just means *someone* who does *something*: in this case, you manage an equities portfolio.
2. You take **actions**, such as buying and selling stock.
3. In buying and selling stock, you are interacting with an **environment**, where the environment is the stock market or, more broadly, the financial system as a whole.
4. Your buy and sell decisions are based on (your **observations** of) the **state** of the market or of the financial system.
5. Managing the portfolio will bring a **reward**: a positive reward if you do well, a "negative reward" if you do badly.
6. Having traded for any amount of time, your experience teaches you how to trade. In psychology, being rewarded for doing a good job is called positive reinforcement. You are learning through trial and error, and this is a way to think about **reinforcement** **learning **at a very high level.

You might already be able to imagine how this simple RL model can be used to describe almost anything you might do. Watch this required reading video for another description of this RL model, applied to a self-driving car.

Figure 1: Diagram of Reinforcement Learning Model



Source: Vogiatzis, Antonios. *Reinforcement Learning for Financial Portfolio Management*. Feb 2019. Technical University of Crete, MS thesis. https://dias.library.tuc.gr/view/81140.

2. The RL Value Functions and the Optimal Policy: The Bellman Equation

Other important elements of RL include the policy and value function. A **policy** is just what you do and when you do it; more technically, a policy is a mapping of actions to states. In the example above, you might have a policy based on technical indicators: Maybe you have a policy of buying a stock if its price has an upward trend and positive momentum. If the "state" (the trend and momentum) of the stock price is upward and positive, then the "action" is to buy. Or if your policy is based on macroeconomic indicators: You sell bank stocks if you believe interest rates are going down. In that case, the state includes decreasing interest rates, and the action is to sell bank stocks. In RL, the agent will learn its own policy: It will learn how to use the indicators that are provided, or it will learn how to use indicators that it learns through trial and error.

When we refer to *the* value function, we are actually talking about two value functions: the state-value function and the action-value function.

- **State-value function**: Technically and specifically, "the value function of a state s under a policy \pi, denoted v(s), is the expected return when starting in s and following \pi thereafter" (Sutton and Barto 58). With that said, a working definition to get you started follows: "Roughly speaking, the value of a state is the total amount of reward an agent can expect to accumulate over the future, starting from that state" (Sutton and Barto 58).
- **Action-value function**: Technically, "we define the value of taking action a in state s under a policy \pi, denoted q(s, a), as the expected return starting from s, taking the action a, and thereafter following policy \pi" (Sutton and Barto 58). To understand this technical definition consider that the "value of an action depends on the expected next reward and the expected sum of the remaining rewards" (Sutton and Barto 62).

This required reading ten-minute video does a great job of explaining how these two functions are intimately related via the Bellman equation to each other and to the optimal policy. According to the video, the **Bellman equation** is "the cornerstone of all" RL (Körding & Ungar). Indeed, the state-value function depends on the action-value function, and vice-versa. This may sound problematic at first, but in fact it just means that we use the two functions recursively to find the value of actions and states. When we know which actions in which states lead to the highest value, we have our *optimal* policy.

To put this another way, keep in mind that the optimal policy maximizes the agent's *cumulative* reward over a multi-step horizon. The agent seeks the *action* with the highest value (according to the action-value function defined above) as well as the *state* with the highest value (according to the state-value function above), and this information is codified in the optimal policy. Such a policy does not simply dictate the action with the highest reward for that step. The optimal policy must also account for the next state (that is, the state that results from this action) and the probability of the actions, rewards, and states that are available from that next state.

3. Exploration and Exploitation

In learning the optimal policy, the RL agent must make a trade-off at each decision step: Should it take the action that it currently calculates will lead to the highest value or cumulative reward, or should it try different actions to see what reward results? One scholar compares this decision to making your favorite pizza or trying different kinds of pizza and possibly finding one that you like better (Meier). The following puts this dilemma a bit more formally:

"To obtain a lot of reward, a reinforcement learning agent must prefer actions that it has tried in the past and found to be effective in producing reward. But to discover such actions, it has to try actions that it has not selected before. The agent has to *exploit* what it has already experienced in order to obtain reward, but it also has to *explore* in order to make better action selections in the future" (Sutton 3).

Certainly, at the very beginning, the first step is exploratory: The agent has no information about the environment to exploit and therefore must explore. For example, if you have never had pizza before, maybe you try *any* combination of toppings to start and adjust accordingly when making subsequent pizzas.

4. The Policy Gradient Method

Thus far in these lesson notes, we have discussed the policy as a linear mapping, like a table, from state to action based on the optimal value as calculated with the value functions. Now consider instead a policy that encodes this same information in a large set of parameters—say, for example, the weight and bias parameters of a neural net. This would allow the agent to learn a much more complex and, perhaps, better policy than the tabular policy we were imagining before. Recalling our first example in this lesson, the portfolio manager agent could learn how to trade in a more complex market.

This leads us to the procedure used to update the agent's policy—in search of the potentially complex *optimal* policy. Just as we can use gradient descent to optimize the parameters of a neural net with respect to a loss function, we can use gradients to optimize the vector of policy parameters with respect to the value function. A method that updates its policy parameters in this way is called a **policy gradient method** (PGM). Similar to standard gradient descent, the policy parameters are updated in the direction of the gradient (where the slope with respect to value is the steepest) in an effort to find the optimal policy where the value is a maximum—ideally, a global maximum and not a local maximum.

5. Types of Agents in Reinforcement Learning

To keep things simple, we have been assuming only one type of agent, but in fact, there are many types of agents. The type of agent implies how the agent constructs its policy, which really is the mechanics of the RL process. These are referenced and described in more detail in the readings, but we provide these high-level descriptions to help prepare you for the more technical definitions you will discover (Vogiatzis 11):

- "Value-Based Agent, the agent will evaluate all the states in the state space, and the policy will be kind of implicit, i.e. the value function tells the agent how well each action in a given state is and the agent selects the best.
- Policy-Based Agent, instead of representing the value function inside the agent, we explicitly represent the policy. The agent searches for the optimum action value function so it can act optimally.
- Actor-Critic Agent, this agent is a value-based and policy-based agent. It's an agent that stores both the policy and each state's reward.
- Model-Based Agent, the agent tries to build a model of how the environment works, and then plan to get the best possible behavior.
- Model-Free Agent, here the agent does not try to understand the environment, i.e., we do not try to build the dynamics. Instead, we go directly to the policy and/or value function. We just see experience and try to figure out a policy of how to behave optimally to get the best possible rewards."

6. Conclusion

In this lesson, we reintroduced reinforcement learning in the context of portfolio management and optimization. We build on much of what we have covered in this course to date as well as recalling certain key concepts from the Machine Learning course. In the next lesson, we apply what we now know about reinforcement learning to more hands-on examples of portfolio management and trading.

References

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